# Integrative Industry Synthesis

AI-based Traffic Sign Recognition & Safety Validation System, Capstone Project 7

## Overview and Industry Context

Advanced driver-assistance systems depend on perception models that can misread a road sign and still behave safely when they do. A traffic sign classifier is never going to be perfect: signs are weathered, partly occluded, photographed at odd angles, or simply rare enough that a model sees too few examples of them during training. The industry question this project addresses is not "can a classifier read signs accurately," which Project 4 already answered with a working CNN, but "what does a deployment-realistic system need around that classifier so a wrong reading does not quietly become a wrong driving decision." That question only shows up once the pieces are combined. A classifier report can state an accuracy number and stop there; a system that is actually responsible for feeding a downstream decision needs a plan for its own uncertainty, a way to get better at the cases it is worst at, and a layer that catches the cases it should not act on alone.

This project integrates three prior components into one new pipeline: the CNN classifier from Project 4, a generative model extending the VAE approach from Project 5, and the agentic safety layer from Project 6. The result is framed as a single system, an AI-based Traffic Sign Recognition & Safety Validation System.

## Integration Rationale

Project 4's CNN architecture was trained on the German Traffic Sign Recognition Benchmark (GTSRB; Stallkamp, Schlipsing, Salmen, & Igel, 2012), but Project 4 never saved its weights, so this project reproduces that run (same architecture, hyperparameters, seed), as Project 6 did. That reproduction reaches 84.3% test accuracy, a few points above Project 4's own 82.0%, a gap traced to a train/validation split difference: Project 4's notebook drew two unrelated random samples before shuffling the split with that same generator, while this reproduction shuffles fresh. It is weaker on rare classes regardless: this project's own per-class analysis of that checkpoint found ten classes below roughly 69% accuracy, several well below 50%, almost all with only 60 to 150 test examples. Project 5 showed that a VAE can generate synthetic sign crops, but could not condition generation on a specific class label, because its dataset (Mapillary Traffic Sign Dataset) spread about 400 categories across roughly 1,700 usable crops, far too sparse to learn a reliable per-class generator. GTSRB does not have that sparsity problem. Its 43 classes have thousands of examples

each, which makes conditioning on label viable for the first time in this project series. A class-conditional VAE (extending Sohn, Lee, & Yan's 2015 formulation, and Kingma and Welling's 2014 VAE foundations that Project 5 already used) trained on GTSRB can generate synthetic crops for a chosen class, which Project 5's model structurally could not do. Those synthetic crops can be weighted toward the classes Project 4's classifier is weakest on, which is a far more targeted augmentation strategy than either doing nothing or applying a generic rotation and color-jitter transform uniformly. The resulting, hopefully improved, classifier is then wrapped in Project 6's safety agent unmodified, so the question becomes whether a better classifier changes the agent's real behavior, not just its own standalone accuracy number. Each project contributes something load-bearing to the next stage; none of the three is included because it happened to also be about traffic signs.

## System Design and Technical Decisions

The pipeline has three stages (Figure 1). Stage A trains a class-conditional VAE on the GTSRB training split: the architecture mirrors Project 5's convolutional encoder/decoder, but a one-hot class label is concatenated into both the encoder's bottleneck and the decoder's input, so a label can be supplied at generation time. It trained for 150 epochs on this project's laptop (an M-series Mac, MPS backend) in about eleven minutes, producing legible, class-consistent speed-limit digits and correctly differentiated sign shapes, a clear improvement on Project 5's generation quality that came from conditioning and GTSRB's larger per-class sample count.

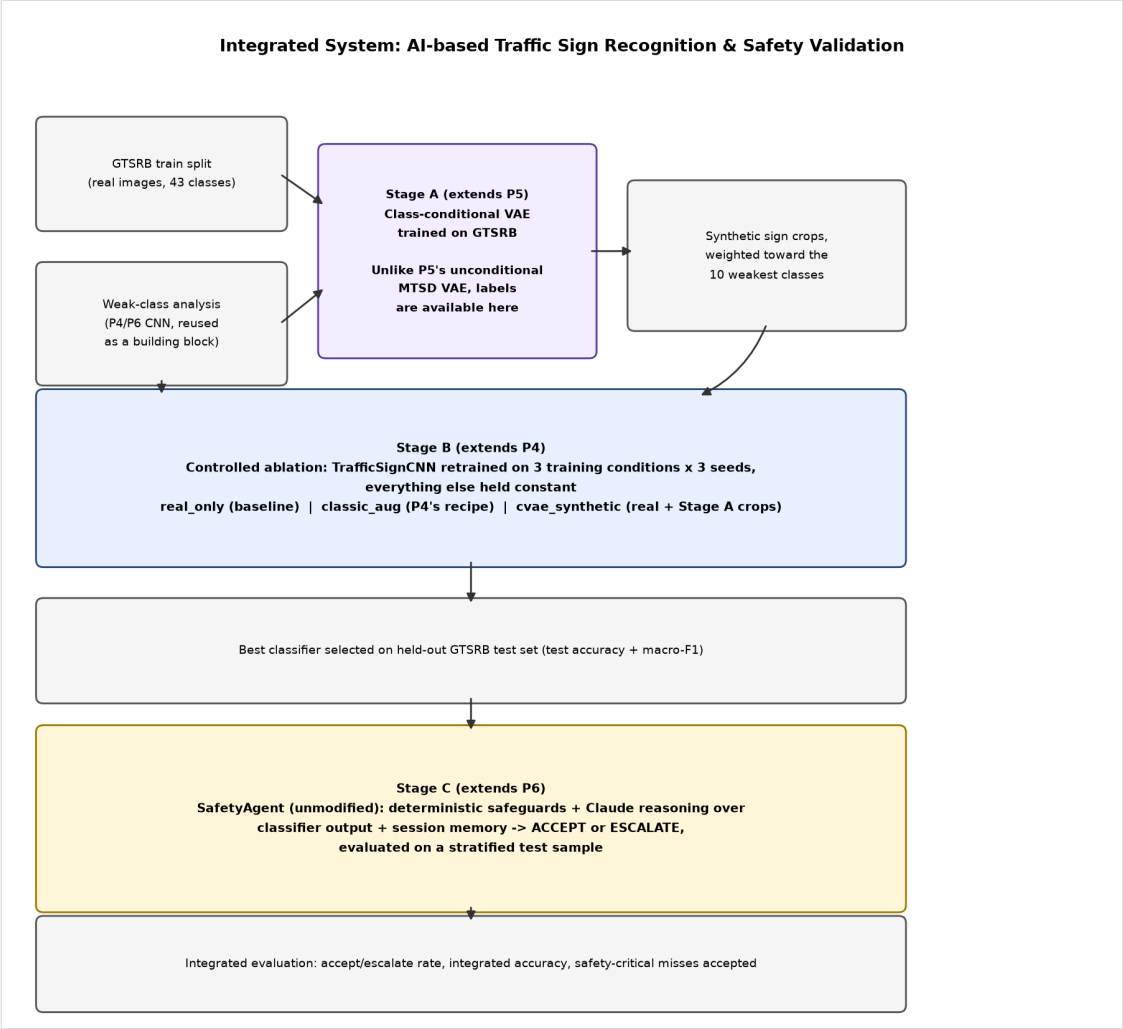



Figure 1. The three-stage integrated pipeline: Stage A (extends Project 5) generates class-conditional synthetic crops weighted toward the weakest classes, Stage B (extends Project 4) retrains the classifier under a controlled ablation, Stage C (extends Project 6) wraps the best classifier in the unmodified safety agent.




Figure 2. Class-conditional VAE samples. Columns 0-7 (speed-limit and related classes) repeat across four rows, showing consistent per-class conditioning: legible digits, correct circular red rims for speed limits, and a correctly differentiated grayscale diagonal-slash shape for "end of limit" signs.

Stage B is a controlled ablation, not a single comparison. The Project 4 CNN architecture was retrained under three training-data conditions, three random seeds each, holding the architecture, epoch count, batch size, optimizer, learning rate, and train/validation/test split identical across all nine runs: real data only (the original baseline), real data with Project 4's existing rotation/affine/color-jitter augmentation, and real data combined with Stage A's synthetic crops, weighted toward the ten weakest classes. The synthetic-augmentation condition won on both metrics (mean test accuracy 84.5% versus 83.2% for the baseline and 80.3% for classic augmentation; mean macro-F1 0.794 versus 0.760 and 0.749). The macro-F1 gain was proportionally larger than the accuracy gain, consistent with the augmentation being concentrated on rare classes that weigh more heavily in a macro-averaged metric. Classic augmentation actually underperformed the untouched baseline here, which matches what Project 4 itself originally found with the same recipe and architecture.

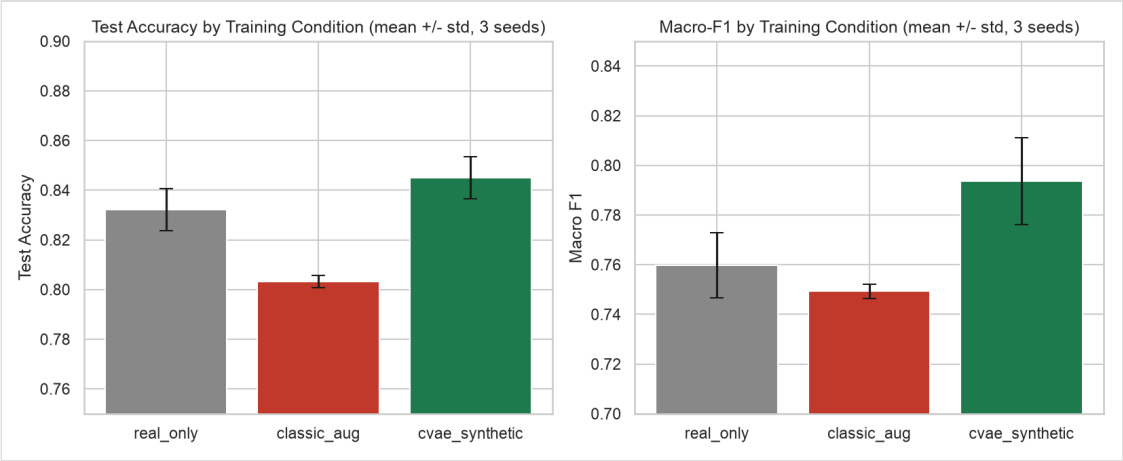



Figure 3. Test accuracy and macro-F1 by training condition, mean and standard deviation across 3 seeds. The synthetic-augmentation condition leads on both metrics; classic augmentation underperforms the baseline.

Stage C wraps the two most relevant classifiers, the real-data-only baseline and the synthetic-augmented winner, in Project 6's agent (deterministic confidence/margin/safety-critical thresholds, with a Claude reasoning step over the classifier's structured output and short-term session memory) without changing a single line of the agent itself. Both were evaluated over the same stratified sample of the GTSRB test set, ten cases per class, 430 cases per classifier, so every class including the rare and safety-critical ones gets meaningful representation without 430 times the cost of scanning the full 12,630-image test set with a live LLM call on every case.

## Ethical Considerations and Responsible AI

The clearest risk in this system is a false negative that looks like a success: a confidently wrong classification that clears every threshold and gets auto-accepted. A concrete instance surfaced directly in this project's own run. Three true "Speed limit (20km/h)" test images were misread as "Speed limit (30km/h)" by the baseline classifier. Two of the three were read at 96% and over 99% confidence and auto-accepted, because nothing about a specific, confident, wrong prediction trips any threshold designed to catch uncertainty; the third was read at only 62% confidence and was correctly escalated by the same baseline agent. The synthetic-augmented classifier still misread all three of these same images as "Speed limit (30km/h)", so the underlying prediction error was not fixed, but its confidence on all three dropped (81%, 57%, 57%), which was enough to push every one of them past the safety-critical escalation threshold instead of being auto-accepted. That is a real, if partial, improvement, but it is a change in calibration, not a fix to the structural gap: every safeguard in this design checks a property of the predicted class, because the agent has no access to

ground truth at inference time, so a confident misclassification can always slip through regardless of how good the upstream classifier gets. The full evaluation confirms this at scale: the synthetic-augmented classifier's agent-wrapped system still auto-accepted five safety-critical misclassifications out of 430 sampled cases, the identical count as the baseline.

Beyond that specific failure mode, dataset bias is a direct carryover risk. GTSRB's own class imbalance, the same rare signs that drove this project's augmentation choices, means the system has seen real-world examples of some sign types an order of magnitude more often than others, and all of that data comes from one country's roads and sign conventions. Deploying a system trained this way anywhere else, or on different weather and lighting conditions than GTSRB's collection vehicle encountered, would need a fresh evaluation, not an assumption that these results transfer. There is also an accountability question specific to combining a generative model with a safety-critical classifier: if a synthetic-crop artifact in the augmented training data caused a specific new misclassification pattern, responsibility would sit awkwardly between the generative model's output, the classifier's training process, and the safety agent's thresholds, none of which were individually "wrong" in isolation.

## Evaluation and Reflection

Does the integrated system meet its goal? Partially. The classifier genuinely improved from targeted synthetic augmentation, by a margin that held up across three seeds, not a single lucky run. That improvement carried through to the agent-wrapped system: accuracy among auto-accepted cases rose from 89.2% to 91.9%, and the count of wrong auto-accepts across the full sample dropped from 32 to 23 out of 430 cases, a real reduction in the volume of silent errors a human reviewer would never see. The system did not, however, close the specific safety-critical failure mode that matters most: the count of safety-critical misclassifications that still slipped through auto-accept stayed at five in both configurations. That is the central limitation of this integrated approach, better training data improves general reliability but does not, by itself, fix a structural blind spot in how the safety layer makes its decisions. A system this complex also trades interpretability for performance: three stages, each with its own hyperparameters and sampling choices, are harder to audit end to end than any one component alone, even though each individual stage is simpler than the whole.




Figure 4. Integrated-system evaluation over a stratified 430-case test sample per classifier. The synthetic-augmented classifier's agent-wrapped system is more accurate among auto-accepted cases, but the count of safety-critical misclassifications that still clear auto-accept is unchanged.

Project 4's classic augmentation recipe, rotation, affine jitter, and color jitter, underperformed doing nothing at all in this project's own re-run, not just in the original Project 4 report. That consistency across two independent training runs, months apart, on the same architecture, is useful evidence that the result is not noise, and it reinforces the actual finding of this project: augmentation strategy matters more than whether augmentation happens at all, and a targeted, class-aware synthetic strategy outperformed a generic geometric one on this specific architecture and dataset.

## Professional Relevance

Real AI deployments are rarely a single model in isolation; they are pipelines where one component's output becomes another's input, and where the interesting failures happen at the seams rather than inside any one piece. Building and evaluating this system end to end required reasoning about what each component could and could not supply to the next one: GTSRB's class imbalance drove Stage A's generation targets, the resulting classifier's actual test-set behavior determined what Stage C's agent would see, and the agent's unmodified logic made it possible to isolate whether an upstream change (the classifier) actually mattered downstream (the agent's decisions), which is exactly the kind of question a team shipping a real perception-plus-safety pipeline has to answer before a release, not after.

## References


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